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~ PRDB ~

★ A random variable is a function that associates a real number with each element in the sample space. Represented by X .

1 e.g. Consider sample space of tossing two coins $\rightarrow S = \{HH, HT, TH, TT\}$

Let X be "random variable", that in this case represents "No. of Head".

Then $X = \{0, 1, 2\} \rightarrow$

Sample Space	X
HH	2
HT	1
TH	1
TT	0

! Why random variable?

The sample space gives detailed outcome

But, sometime we don't care about

the exact pattern.

2 e.g. A experiment to "Test three electronic components" may be written as

$S = \{NNN, NND, NDN, DNN, NDD, DND, DDN, DDD\}$ ($N \rightarrow$ Non-Defective)
($D \rightarrow$ Defective)

..... We care about something numerical like:

"How many defective items are there?"

So, we create a random variable. 😊

$X = \{0, 1, 2, 3\}$ i.e. there can be no defective component, 1, 2, or all defective

3 e.g. An urn contains 4 Red balls and 3 black balls. If two balls are drawn without replacement then Let "Y = number of red balls"

$Y = \{0, 1, 2\} \rightarrow$

Sample Space	Y
RR	2
RB	1
BR	1
BB	0

Bernoulli Random Variable (Binary) Sometimes outcomes are just

In such cases the random variable X is defined as,

$$X = \begin{cases} 1, & \text{if defective} \\ 0, & \text{if non-defective} \end{cases}$$

yes / No

Success / Failure

Defective /

Non-defective

[Must read the last page first] (👉 ^o^) 👈

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★ "Types of Random Variable" ~

DRV - Discrete Random Variable if its set of possible outcomes is countable. ★ [Count - "How many"]
In all previous examples so far random variable is Discrete Random Variable.

CRV - Continuous Random Variable if it assume any possible value between certain limit i.e. values from an entire interval of numbers. ★ [Measure - "How much"]
e.g. measured data such as all possible heights, weights, temperature etc.

(3.2) Probability Distribution

Discrete Probability Distribution — before

Whenever a random variable is assigned a value there is definitely gonna be certain probability value of it, i.e. $P(X)$ if random value is X .

The arrangement of each random value with its corresponding probability value is called **Probability Distribution**.

This set of ordered pairs $(x, f(x))$ is also called **probability function**, **probability mass function** of discrete random variable X .

x	$f(x) = P(X=x)$
x_1	$P(X=x_1)$
x_2	$P(X=x_2)$
$\vdots$	$\vdots$
x_n	$P(X=x_n)$

"Discrete Probability Distribution"

★ Properties

1) $0 \leq f(x) \leq 1$

2) $\sum P(x) = 1$

4 e.g Tossing 3 fair coins $S = \{HHT, HHH, HTH, HTT, THH, THT, TTH, TTT\}$
let X = "number of heads"

x	$P\{X=x\}$
0	$1/8$
1	$3/8$
2	$3/8$

0 head $\rightarrow x=0, P\{X=0\} = P(TTT) = 1/8$

$x=1 \Rightarrow P\{X=1\} = P(HTT) + P(THT) + P(THH) = 3/8$

$x=2 \Rightarrow P\{X=2\} = P(HHT) + P(HTH) + P(THH) = 3/8$

Total = $1/8 + 3/8 + 3/8 = 1$

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Prob.

5 e.g A discrete random variable having following probability distribution:

X	0	1	2	3	4	5
P(X)	0	k	0.2	2k	0.3	2k

Find a) Value of k b) $P(X < 3)$ c) $P(X \geq 3)$ d) $P(2 < X < 5)$

a) Since $\sum P(X) = 1 \Rightarrow P(0) + P(1) + P(2) + P(3) + P(4) + P(5) = 1$
 $\Rightarrow 0 + k + 0.2 + 2k + 0.3 + 2k = 1$
 $\Rightarrow k = 0.1$

b) $P(X < 3) = P(0) + P(1) + P(2) = 0 + k + 0.2 = 0.1 + 0.2 = 0.3$

c) $P(X \geq 3) = P(3) + P(4) + P(5) = 2k + 0.3 + 2k = 0.2 + 0.3 + 0.2 = 0.7$

d) $P(2 < X < 5) = P(3) + P(4) = 2k + 0.3 = 2(0.1) + 0.3 = 0.2 + 0.3 = 0.5$

6 e.g A coin is biased so that the head is 3 times as likely to occur as a tail.

If coin is tossed twice, find the probability distribution for the number of tails.

$P(\text{tail}) + P(\text{head}) = 1 \Rightarrow x + 3x = 1 \Rightarrow x = 1/4$

$P(T) = 1/4, P(H) = 3/4$

X = "number of tails in 2 tosses"

$X = \{0, 1, 2\}$

i) $X=0$ $P(X=0) = P(HH) = (3/4)(3/4) = 9/16$

ii) $X=1$ $P(X=1) = P(HT) + P(TH) = (3/4)(1/4) + (1/4)(3/4) = 3/8$

iii) $X=2$ $P(X=2) = P(TT) = (1/4)(1/4) = 1/16$

X	P(X=x)
0	9/16
1	3/8
2	1/16

* Alternative - Binomial Distribution (Later In detail)

Number of trials: $n=2$

Probability of tail: $p = 1/2$ (p is probability of success)

probability of head: $q = 3/4$ (q is probability of failure)

X: "number of tails"

$P(X=x) = \binom{n}{x} p^x q^{n-x}$

$P(0) = \binom{2}{0} (1/4)^0 (3/4)^2 = 9/16$

Similarly $P(1) = 3/8, P(2) = 1/16$

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7 e.g. 20 laptops, 3 are defective. If you have to choose 2, find the probability distribution of no. of defectives

X = "number of defective computers out of 2 chosen computers"

$$X = \{0, 1, 2\} \quad P(X=x) = \frac{\text{Ways to get } x \text{ defectives}}{\text{Total ways to choose 2 laptops}} = \frac{\binom{\text{defective}}{x} \binom{\text{good}}{2-x}}{\binom{\text{total}}{2}}$$

$$f(0) = \frac{\binom{3}{0} \binom{17}{2}}{\binom{20}{2}} = \frac{1 \cdot 136}{190} = \frac{68}{95}$$

$$f(1) = \frac{\binom{3}{1} \binom{17}{1}}{\binom{20}{2}} = \frac{3 \cdot 17}{190} = \frac{51}{190}$$

$$f(2) = \frac{\binom{3}{2} \binom{17}{0}}{\binom{20}{2}} = \frac{3 \cdot 1}{190} = \frac{3}{190}$$

X	$P(X=x)$
0	68/95
1	51/190
2	3/190
Total	1

8 e.g. A car agency sells 50% cars with airbags. Find a formula for the probability distribution of the number of cars with side airbags among next 4 cars sold by agency. (Using Binomial Distribution)

X = "number of cars with airbags"

No. of cars sold = $n = 4$

Probability of having airbags = $p = 50\% = 0.5$

Probability of not having airbags = $q = 1 - 0.5 = 0.5$

★

Total Probability = Success Probability + Failure Probability = 1

$p + q = 1 \Rightarrow q = 1 - p$

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \binom{4}{x} (0.5)^x (0.5)^{4-x}$$

$$\therefore (0.5)^x (0.5)^4 = 0.5^4 = \frac{1}{16}$$

$$f(x) = \binom{4}{x} \frac{1}{16}$$

$$X = \{0, 1, 2, 3, 4\}$$

X	0	1	2	3	4
$P(X)$	$1/16$	$1/4$	$3/8$	$1/4$	$1/16$

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(link is last page)

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★ Cumulative Distribution Function $F_X(x) = P(X \leq x) = \sum_{y \leq x} P(y)$

9 e.g. Rolling of six-sided die.

X	0	1	2	3	4	5	6
P(X)	0	1/6	1/6	1/6	1/6	1/6	1/6

$$F_X(0) = 0$$

$$F_X(1) = P(0) + P(1) = 0 + 1/6 = 1/6$$

$$F_X(2) = P(0) + P(1) + P(2) = 0 + 1/6 + 1/6 = 2/6$$

$$F_X(3) = P(0) + P(1) + P(2) + P(3) = 2/6 + 1/6 = 3/6$$

$$F_X(4) = 4/6 \quad F_X(5) = 5/6 \quad F_X(6) = 6/6 = 1$$

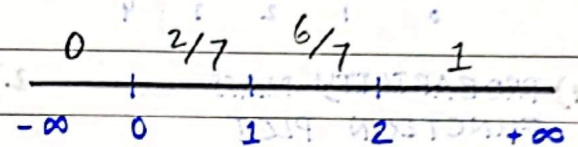
10 e.g. Find Cumulative Distribution Function of following probability distribution.

X	0	1	2
P(X)	2/7	4/7	1/7

$$f(0) = 2/7$$

$$f(1) = 2/7 + 4/7 = 6/7$$

$$f(2) = 6/7 + 1/7 = 1$$



$$f(x) = \begin{cases} 0 & ; x < 0 \\ 2/7 & ; 0 \leq x < 1 \\ 6/7 & ; 1 \leq x < 2 \\ 1 & ; x \geq 2 \end{cases}$$

11 e.g. (DIY) Probability Distribution of X, the no. of imperfections per 10 meters of synthetic fabric in continuous rolls of uniform is given by

X	0	1	2	3	4
P(X)	0.41	0.37	0.16	0.05	0.01

Ans: $F(x) = \begin{cases} 0 & ; x < 0 \\ 0.41 & ; 0 \leq x < 1 \\ 0.78 & ; 1 \leq x < 2 \\ 0.94 & ; 2 \leq x < 3 \\ 0.99 & ; 3 \leq x < 4 \\ 1 & ; x \geq 4 \end{cases}$

X	0	1	2	3	4
P(X)	1/16	2/4	3/8	1/4	2/16

DPD/PMD for car agency

(from last example of previous page)

$$F(0) = 1/16$$

$$F(1) = 1/16 + 1/4 = 5/16$$

$$F(2) = 5/16 + 3/8 = 11/16$$

$$F(3) = 11/16 + 1/4 = 15/16$$

$$F(4) = 15/16 + 1/16 = 1$$

$$F(x) = \begin{cases} 0 & ; x < 0 \\ 1/16 & ; 0 \leq x < 1 \\ 5/16 & ; 1 \leq x < 2 \\ 11/16 & ; 2 \leq x < 3 \\ 15/16 & ; 3 \leq x < 4 \\ 1 & ; x \geq 4 \end{cases}$$

(?) Using F(X) verify $f(2) = 3/8$

$$f(2) = F(2) - F(1) = 3/8$$

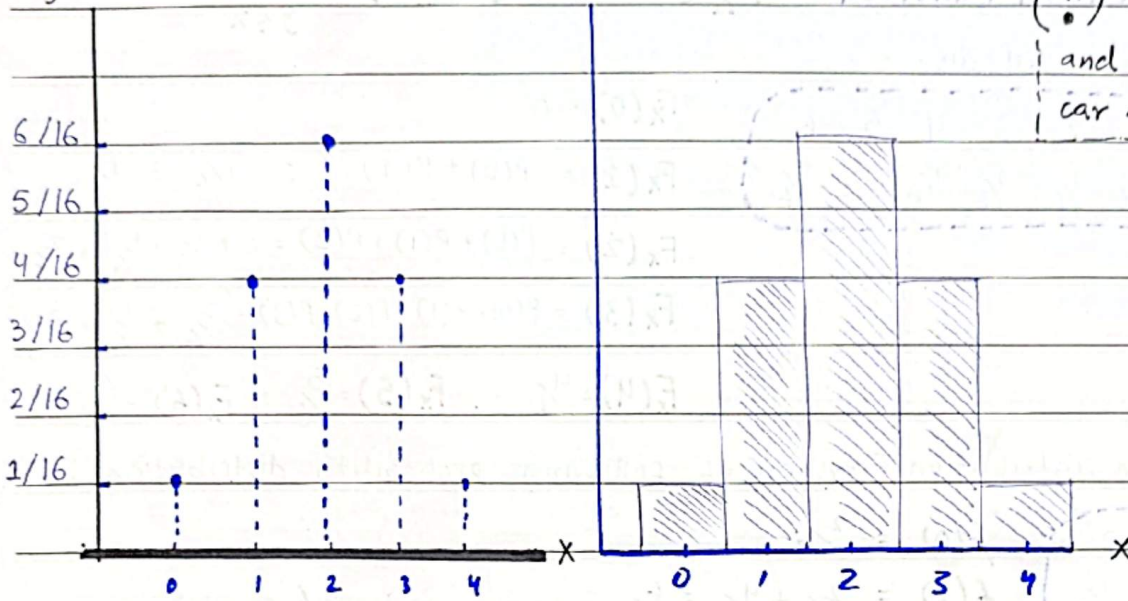
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Probability

$f(x)$ or $P(x)$

$f(x)$

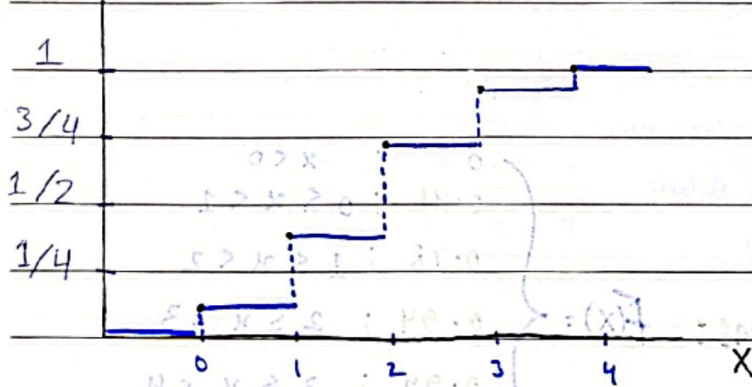
(!) all graphs - 1), 2) and 3) are for above car agency example



1) PROBABILITY MASS FUNCTION PLOT

2) PROBABILITY HISTOGRAM

$F(x)$



3) DISCRETE CUMULATIVE DISTRIBUTION FUNCTION

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★ Continuous Probability Distribution - If random variables are continuous then probability distribution is called continuous probability distribution.

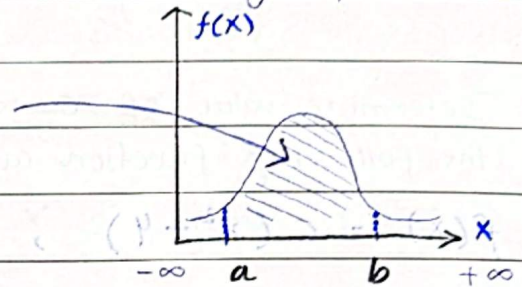
→ such distribution can't be presented in tabular form, it can be stated as a formula.

That function $f(x)$ is called probability density function (PDF).

Let's suppose that you are asked to find "probability of all random variables between 'a' and 'b', then

$P(a \leq x \leq b) = \text{Area under curve}$

$$P(a \leq x \leq b) = \int_a^b f(x) dx$$



★ Property of PDF

$$1) \int_{-\infty}^{+\infty} f(x) dx = 1$$

13 Q9 Check whether the given function is PDF or not.

$$1) f(x) = x/2 ; 0 \leq x \leq 2 \quad 2) f(x) = 0 ; x < 2$$

Since, we know $\int_{-\infty}^{+\infty} f(x) dx = 1$

$$= \frac{1}{18} (3+2x) ; 2 \leq x \leq 4$$

$$= 0 ; x > 4$$

$$\int_0^2 \frac{x}{2} dx = \frac{1}{2} \left[\frac{x^2}{2} \right]_0^2 = \frac{1}{4} [4 - 0] = \frac{4}{4} = 1$$

As, $\int_0^2 \frac{x}{2} dx = 1$. So it is probability density function.

$$2) \int_{-\infty}^{+\infty} f(x) dx = \int_{-\infty}^2 0 dx + \int_2^4 \frac{1}{18} (3+2x) dx + \int_4^{\infty} 0 dx$$

$$= 0 + \frac{1}{18} [3x + x^2]_2^4 + 0$$

$$= \frac{1}{18} [(12+16) - (6+4)]$$

$$= \frac{1}{18} [12+6] = \frac{18}{18}$$

$$= 1 \text{ hence it is a PDF.}$$

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14 e.g. $f(x) = \begin{cases} x^2/3 & ; -1 < x < 2 \\ 0 & ; \text{elsewhere} \end{cases}$ a) Verify if $f(x)$ is PDF b) Find $P(0 < X \leq 1)$

a) $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{-1} 0 dx + \int_{-1}^2 \frac{x^2}{3} dx + \int_2^{\infty} 0 dx$

$= 0 + \frac{1}{3} \left[\frac{x^3}{3} \right]_{-1}^2 + 0$

$= \frac{1}{9} (8 - (-1))$

$= \frac{9}{9} = 1$

b)

$P(0 < x \leq 1) = \int_0^1 \frac{x^2}{3} dx$

$= \frac{1}{9} [x^3]_0^1$

$= \frac{1}{9}$

15 e.g. The shelf life in days, for bottles of a certain prescribed medicine is

Random variable X having DF

$f(x) = \begin{cases} \frac{20000}{(x+100)^3} & ; x > 0 \\ 0 & ; \text{elsewhere} \end{cases}$

Find probability that a bottle of medicine will have shelf life of

a) atleast 200 days

$P(X \geq 200) = \int_{200}^{\infty} \frac{20000}{(x+100)^3} dx$

$= 20000 \int_{200}^{\infty} (x+100)^{-3} dx$

(let $u = x+100$)

$= 20000 \int_{200}^{\infty} u^{-3} du$

$= 20000 \left[\frac{u^{-2}}{-2} \right]_{200}^{\infty} = 20000 \left[-\frac{1}{2(x+100)^2} \right]_{200}^{\infty}$

$= \frac{20000}{-2} \left[\frac{1}{\infty^2} - \frac{1}{(300)^2} \right]$

$= 10000 \left(\frac{1}{90000} \right) = \frac{1}{9}$

b) anywhere from 80 to 120 days

$P(80 \leq x \leq 120) = \int_{80}^{120} \frac{20000}{(x+100)^3} dx$

$= 20000 \int_{80}^{120} u^{-3} du$

$= \frac{20000}{-2} \left[\frac{1}{(x+100)^2} \right]_{80}^{120}$

$= -10000 \left[\frac{1}{(220)^2} - \frac{1}{(180)^2} \right]$

$= -10000 \left(\frac{1}{48400} - \frac{1}{32400} \right)$

$= \frac{160000000}{1568160000} = 0.102$

16 e.g. $f(x) = \begin{cases} k\sqrt{x} & ; 0 < x < 1 \\ 0 & ; \end{cases}$

a) Find k $1 = \int_0^1 k\sqrt{x} \Rightarrow \frac{2k}{3} \left[x^{3/2} \right]_0^1 \Rightarrow \frac{2k}{3} = 1 \Rightarrow k = \frac{3}{2}$

b) $P(0.3 < x < 0.6) = \frac{2}{3} \left(\frac{3}{2} \right) \left[x^{3/2} \right]_{0.3}^{0.6} = (0.6)^{3/2} - (0.3)^{3/2} = 0.3004$

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PROBABILITY

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★ Cumulative Distribution Function $F(x)$ of continuous random variable X with density function $f(x)$ is $F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$

17 e.g $f(x) = \begin{cases} 6x(1-x) & ; 0 \leq x \leq 1 \\ 0 & ; \text{elsewhere} \end{cases}$ Find CDF for X .

$$\therefore F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$

$$= \int_0^x 6x(1-x) dx = \int_0^x (6x - 6x^2) dx$$

$$= \left[\frac{6x^2}{2} - \frac{6x^3}{3} \right]_0^x = 3x^2 - 2x^3$$

$$F(x) = \begin{cases} 0 & ; x < 0 \\ 3x^2 - 2x^3 & ; 0 \leq x \leq 1 \\ 1 & ; x > 1 \end{cases}$$

18 e.g Find value of K for the PDF $f(x) = \begin{cases} Kx^2 & ; 0 \leq x \leq 3 \\ 0 & ; \text{elsewhere} \end{cases}$ and compute $P(1 \leq x \leq 2)$. Also find CDF.

$$(\text{For } K) \rightarrow \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^3 Kx^2 dx = \frac{K}{3} [x^3]_0^3 = 9K \Rightarrow 9K = 1 \Rightarrow K = 1/9$$

$$(\text{For } P(1 \leq x \leq 2)) \rightarrow \int_1^2 Kx^2 dx$$

$$\Rightarrow \int_1^2 \frac{1}{9} x^2 dx \Rightarrow \frac{1}{9} \cdot \frac{1}{3} [x^3]_1^2 = \frac{1}{27} [8 - 1] = \frac{7}{27}$$

$$(\text{For CDF}) \rightarrow F(x) = \int_{-\infty}^x f(x) dx$$

$$\Rightarrow \int_0^x \frac{1}{9} x^2 dx = \frac{1}{9} \cdot \frac{1}{3} [x^3]_0^x = \frac{x^3}{27}$$

$$\Rightarrow F(x) = \begin{cases} 0 & ; x < 0 \\ \frac{x^3}{27} & ; 0 \leq x \leq 3 \\ 1 & ; x \geq 3 \end{cases}$$

19 e.g CDF of X is,

$$F(x) = \begin{cases} 0 & ; x \leq 1 \\ (x-1)^4/16 & ; 1 \leq x \leq 3 \\ 1 & ; x > 3 \end{cases}$$

find PDF and $P(2 < x < 3)$

$$\star f(x) = \frac{d}{dx} F(x)$$

$$f(x) = \begin{cases} \frac{d}{dx} 0 & \\ \frac{d}{dx} \frac{(x-1)^4}{16} & \\ \frac{d}{dx} 1 & \end{cases} \Rightarrow f(x) = \begin{cases} 0 & ; x \leq 1 \\ \frac{1}{4}(x-1)^3 & ; 1 \leq x \leq 3 \\ 0 & ; x > 3 \end{cases}$$

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$$\begin{aligned} \text{(For } P(2 < X \leq 3)) &\rightarrow \int_2^3 f(x) dx \\ \Rightarrow \int_2^3 \frac{1}{4} (x-1)^3 dx &= \frac{1}{4} \int_2^3 (x-1)^3 dx \\ &= \frac{1}{4} \left[\frac{(x-1)^4}{4} \right]_2^3 = \frac{1}{16} [16 - 1] = \frac{15}{16} \end{aligned}$$

20 e.g $f(x) = \begin{cases} x^2/3 & ; -1 < x < 2 \\ 0 & ; \text{otherwise} \end{cases}$ find $F(x)$ and $P(0 < X \leq 1)$

$$\text{(For } F(x)) \rightarrow F(x) = \int_{-\infty}^x f(x) dx$$

$$\Rightarrow F(x) = \int_{-1}^x \frac{x^2}{3} dx = \frac{1}{3} \left[\frac{x^3}{3} \right]_{-1}^x = \frac{1}{9} [x^3 + 1] = \frac{x^3 + 1}{9}$$

$$F(x) = \begin{cases} 0 & ; x < -1 \\ \frac{x^3 + 1}{9} & ; -1 \leq x < 2 \\ 1 & ; x \geq 2 \end{cases}$$

$$\text{(For } P(0 < X \leq 1)) \rightarrow \int_0^1 \frac{x^2}{3} dx$$

$$\Rightarrow \frac{1}{3} \int_0^1 x^2 dx = \frac{1}{3} \cdot \frac{1}{3} [x^3]_0^1 = \frac{1}{9} [1] = \frac{1}{9}$$

$$\begin{aligned} &0 < x < 1 \\ &1 \leq x < 2 \\ &x \geq 2 \end{aligned}$$

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Probability

21 e.g Total number of hours, measured in unit of 100 hours, that a family runs a vacuum cleaner over a period of one year is a continuous RV X that has density function

$$f(x) = \begin{cases} x & ; 0 < x < 1 \\ 2-x & ; 1 \leq x < 2 \\ 0 & ; \text{otherwise} \end{cases}$$

Find probability for

a) less than 120 hours. ($120 \text{ hr} \equiv 1.2$)

b) between 50 and 100 hours. ($0.5 - 1$)

$$\begin{aligned} \text{a) } \int_0^{1.2} f(x) dx &= \int_0^1 x dx + \int_1^{1.2} (2-x) dx = \frac{1}{2} [x^2]_0^1 + \left[2x - \frac{x^2}{2} \right]_1^{1.2} \\ &= \frac{1}{2} + \left[(2 \cdot 1.2 - \frac{1.44}{2}) - (2 - 0.5) \right] = \frac{1}{2} + [1.68 - 1.5] \\ &= \frac{1}{2} + 0.18 = 0.68 = P(X < 1.2) \end{aligned}$$

$$\begin{aligned} \text{b) } \int_{0.5}^1 f(x) dx &= \int_{0.5}^1 x dx = \frac{1}{2} [x^2]_{0.5}^1 = \frac{1}{2} [1 - 0.25] = 0.5 - 0.125 \\ &= 0.375 = P(0.5 < X < 1) \end{aligned}$$

22 e.g

$$f(x) = \begin{cases} \frac{2(x+2)}{5} & ; 0 < x < 1 \\ 0 & ; \text{otherwise} \end{cases}$$

a) Show that $P(0 < X < 1) = 1$

b) $P(\frac{1}{4} < X < \frac{1}{2})$

$$\begin{aligned} \text{a) } P(0 < X < 1) &= \int_0^1 \frac{2(x+2)}{5} dx = \frac{2}{5} \int_0^1 (x+2) dx = \frac{2}{5} \left[\frac{x^2}{2} + 2x \right]_0^1 \\ &= \frac{2}{5} \left[\frac{1}{2} + 2 \right] = \frac{2}{5} \cdot \frac{5}{2} = 1 \quad \square \end{aligned}$$

$$\begin{aligned} \text{b) } P(\frac{1}{4} < X < \frac{1}{2}) &= \int_{\frac{1}{4}}^{\frac{1}{2}} \frac{2(x+2)}{5} dx = \frac{2}{5} \int_{\frac{1}{4}}^{\frac{1}{2}} (x+2) dx = \frac{2}{5} \left[\frac{x^2}{2} + 2x \right]_{\frac{1}{4}}^{\frac{1}{2}} \\ &= \frac{2}{5} \left[\left(\frac{1}{4} + 1 \right) - \left(\frac{1}{16} + \frac{1}{2} \right) \right] = \frac{2}{5} \left[\frac{9}{8} - \frac{17}{32} \right] \\ &= \frac{19}{32} \approx 0.2375 \end{aligned}$$

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★ **Joint Probability Distribution** - If X and Y are two discrete random variables, the probability distribution for their simultaneous occurrence can be represented by $f(x, y)$. In discrete case, $f(x, y) = P(X=x, Y=y)$

23e.9 Two ballpoint are selected from a box that have 3 blue, 2 red, and 3 green pens.

$X =$ "No. of blue pen selected"
 $Y =$ "No. of red pen selected"

★ **Properties**

1) $\sum_x \sum_y f(x, y) = 1$

2) $f(x, y) \geq 0$

a) Find joint probability $f(x, y)$

Possible values of $(X, Y) \rightarrow X = \{0, 1, 2\}, Y = \{0, 1, 2\}$

$\rightarrow \{(0, 0), (0, 1), (1, 0), (1, 1), (0, 2), (2, 0)\}$

Both Green Red, Green Blue Green Blue Red Both Red Both Blue

Joint Probability Function $\rightarrow$

Mass

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}$$

$\binom{3}{x} \rightarrow$ choose blue pens

$\binom{2}{y} \rightarrow$ choose red pens

$\binom{3}{2-x-y} \rightarrow$ choose green pens

$$f(0, 0) = \frac{\binom{3}{0} \binom{2}{0} \binom{3}{2}}{\binom{8}{2}} = \frac{1 \cdot 1 \cdot 3}{28} = \frac{3}{28}$$

$$f(0, 1) = \frac{\binom{3}{0} \binom{2}{1} \binom{3}{1}}{28} = \frac{3}{14}$$

$$f(1, 0) = \frac{\binom{3}{1} \binom{2}{0} \binom{3}{1}}{28} = \frac{9}{28}$$

$$f(1, 1) = \frac{\binom{3}{1} \binom{2}{1} \binom{3}{0}}{28} = \frac{3}{14}$$

$$f(0, 2) = \frac{\binom{3}{0} \binom{2}{2} \binom{3}{0}}{28} = \frac{1}{28}$$

$$f(2, 0) = \frac{\binom{3}{2} \binom{2}{0} \binom{3}{0}}{28} = \frac{3}{28}$$

		x			Row Total
		$f(x, y)$	0	1	2
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Total		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

(Table 3.1)

24 Example: Find C if $f(x, y) = Cxy$; $x = 1, 2, 3$, $y = 1, 2, 3$

$$\sum_x \sum_y f(x, y) = 1 \Rightarrow f(1, 1) + f(1, 2) + f(1, 3) + f(2, 1) + f(2, 2) + f(2, 3) + f(3, 1) + f(3, 2) + f(3, 3) = 1$$

		x			Row Total
		$f(x, y)$	1	2	3
y	1	C	$2C$	$3C$	$6C$
	2	$2C$	$4C$	$6C$	$12C$
	3	$3C$	$6C$	$9C$	$15C$
Col Total		$6C$	$12C$	$15C$	1

$$\Rightarrow C + 2C + 3C + 2C + 4C + 6C + 3C + 6C + 9C = 33C$$

$$\Rightarrow C = \frac{1}{33}$$

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25 Example:

$$f(x, y) = \frac{x+y}{30} \quad \begin{matrix} x = 0, 1, 2, 3 \\ y = 0, 1, 2 \end{matrix}$$

- a) $P(x \leq 2, y = 1) = f(0, 1) + f(1, 1) + f(2, 1) = \frac{1}{30} + \frac{2}{30} + \frac{3}{30} = \frac{6}{30} = \frac{1}{5}$
 b) $P(x > 2, y \leq 1) = f(3, 0) + f(3, 1) = \frac{3}{30} + \frac{4}{30} = \frac{7}{30}$
 c) $P(x+y = 4) = f(2, 2) + f(3, 1) = \frac{4}{30} + \frac{4}{30} = \frac{8}{30} = \frac{4}{15}$

26 Example:

Three cards are drawn without replacement from the 12 face cards (Jacks, Queens, & kings) of 52 playing cards.

X = "Number of kings", Y = "Number of Jacks"

a) Find Joint Probability Distribution

• Total Outcomes ${}^{12}C_3 = 220$

• From Standard Deck: FaceCard = 12

Kings = 4, Jack = 4, Queen = 4

• $f(x, y) = \frac{{}^4C_x {}^4C_y {}^{(3-x-y)}C_{3-x-y}}{220}$

• valid Only when $x, y \geq 0, x+y \leq 3$

$f(0, 0) = \frac{{}^4C_0 {}^4C_0 {}^4C_3}{220} = \frac{4}{220}$

$f(0, 1) = \frac{24}{220}$ $f(1, 0) = \frac{24}{220}$ $f(2, 0) = \frac{24}{220}$

$f(0, 2) = \frac{24}{220}$ $f(1, 1) = \frac{64}{220}$ $f(2, 1) = \frac{24}{220}$

$f(0, 3) = \frac{4}{220}$ $f(1, 2) = \frac{24}{220}$ $f(2, 2) = \frac{4}{220}$

b) $P[(x, y) \in A], \text{ where } A = \{(x, y) \mid x+y \geq 2\}$

$\Rightarrow P(x+y \geq 2) = f(0, 2) + f(0, 3) + f(1, 1) + f(1, 2) + f(2, 0) + f(2, 1) + f(3, 0)$
 $= \frac{(24 + 4 + 64 + 24 + 24 + 24 + 4)}{220} = \frac{144}{220} = \frac{36}{55} \approx 0.6545$

$f(x, y)$	0	1	2	3
0	$\frac{4}{220}$	$\frac{24}{220}$	$\frac{24}{220}$	$\frac{4}{220}$
1	$\frac{24}{220}$	$\frac{64}{220}$	$\frac{24}{220}$	0
2	$\frac{24}{220}$	$\frac{24}{220}$	0	0
3	$\frac{4}{220}$	0	0	0

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★ Marginal Probability Function (for continuous random variable)

$g(x)$ and $h(y)$ are called marginal probability functions,

$$g(x) = \int_{-\infty}^{\infty} f(x,y) dy \quad ; \quad h(y) = \int_{-\infty}^{\infty} f(x,y) dx \quad \text{for CRVs.}$$

27 Example: Find marginal distribution ^{function} of $f(x,y) = \frac{xy^2}{8}$; $0 \leq x \leq 2$, $0 \leq y \leq 2$

$$g(x) = \int_0^2 \frac{xy^2}{8} dy = \frac{1}{8} x \cdot \frac{1}{3} [y^3]_0^2 = \frac{1}{24} x \cdot 8 = \frac{x}{3}; \quad 0 \leq x \leq 2$$

$$h(y) = \int_0^2 \frac{xy^2}{8} dx = \frac{y^2}{8} \left[\frac{x^2}{2} \right]_0^2 = \frac{y^2}{8} \cdot 2 = \frac{y^2}{4}; \quad 0 \leq y \leq 2$$

$$g(x) = \frac{x}{3}, \quad h(y) = \frac{y^2}{4}; \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 2$$

28 Example: Find marginal distribution function of $f(x,y) = \frac{1}{8}(6-x-y)$

$$g(x) = \int_2^4 \frac{1}{8}(6-x-y) dy = \frac{1}{8} \left[6y - xy - \frac{y^2}{2} \right]_2^4; \quad 0 \leq x \leq 2, \quad 2 \leq y \leq 4$$

$$= \frac{1}{8} [(24 - 4x - 8 - 12 + 2x + 2)] = \frac{1}{8} (6 - 2x) = \frac{1}{4} (3 - x)$$

$$h(y) = \int_0^2 \frac{1}{8}(6-x-y) dx = \frac{1}{8} \left[6x - \frac{x^2}{2} - xy \right]_0^2 = \frac{1}{8} [(12 - 2 - 2y)]$$

$$= \frac{1}{8} (10 - 2y) = \frac{1}{4} (5 - y) \Rightarrow g(x) = \frac{1}{4} (3 - x), \quad h(y) = \frac{1}{4} (5 - y)$$

29 Example 3.17: Joint Probability Function

Find $g(x)$ and $h(y)$ for $f(x,y)$ -----

$$f(x,y) = \begin{cases} \frac{2}{5}(2x+3y); & 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$g(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_0^1 \frac{2}{5}(2x+3y) dy = \frac{2}{5} \left[2xy + \frac{3y^2}{2} \right]_0^1 = \frac{2}{5} \left(2x + \frac{3}{2} \right)$$

$$h(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_0^1 \frac{2}{5}(2x+3y) dx = \frac{2}{5} \left[\frac{2x^2}{2} + 3xy \right]_0^1 = \frac{2}{5} (1 + 3y)$$

30 Example: Determine if X and Y are independent.

$$f(x,y) = \begin{cases} 2; & 0 < x \leq y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

If $g(x) \cdot h(y) = f(x,y)$ then X & Y are independent.

$$g(x) = \int_x^1 2 dy = 2(1-x)$$

$$2(1-x) \cdot 2y = 4y(1-x) \neq 2$$

$$h(y) = \int_0^y 2 dx = 2y$$

So, X and Y are NOT independent.

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★ Marginal Probability Function (for Discrete Random Variable)

$g(x)$ and $h(y)$ are called marginal probability functions,

$$g(x) = \sum_{y=1}^Y f(x,y) \quad , \quad h(y) = \sum_{x=1}^X f(x,y) \quad \text{for DRVs.}$$

★ If $g(x) \cdot h(y) = f(x,y)$ then given function $f(x,y)$ is a INDEPENDANT FUNCTION.

31 Example: $f(x,y) = \frac{xy^2}{30}$; $x=1,2,3$; $y=1,2$ Find marginal Distribution Function

$$g(x) = \sum_{y=1}^2 f(x,y) = \sum_{y=1}^2 \frac{xy^2}{30} = \frac{x(1)^2}{30} + \frac{x(2)^2}{30} = \frac{x}{30} + \frac{4x}{30} = \frac{5x}{30}$$

$$h(y) = \sum_{x=1}^3 f(x,y) = \sum_{x=1}^3 \frac{xy^2}{30} = \frac{1(y^2)}{30} + \frac{2(y^2)}{30} + \frac{3(y^2)}{30} = \frac{6y^2}{30} = \frac{y^2}{5}$$

$$f(x,y) = g(x) \cdot h(y) = \left(\frac{5x}{30}\right) \left(\frac{y^2}{5}\right) = \frac{xy^2}{30} \quad \text{Hence } f(x,y) \text{ is independent.}$$

32 Example: JPD Table

		x		
$f(x,y)$		1	2	3
y	1	0.05	0.05	0.10
	3	0.05	0.10	0.35
	5	0.00	0.20	0.10

a) Find Marginal Distribution of X b) Find MD of Y

x	1	2	3
$g(x)$	0.10	0.35	0.55

y	1	3	5
$h(y)$	0.20	0.5	0.3

c) $P(Y=3 | X=2) = \frac{P(X=2, Y=3)}{P(X=2)} = \frac{P(2,3)}{P(2)} = \frac{0.10}{0.35}$

$= \frac{2}{7} \approx 0.286$ [This (c) is Conditional PD. later in detail]

33 Example: Given that: — On two independent tosses $P(\text{Head}) = 40\% = 0.4$

$Z = \text{"No. of Heads on 1st toss"} = \{0, 1\}$ $P(\text{Tail}) = 60\% = 0.6$

$W = \text{"Total No. of Heads"} = \{0, 1, 2\}$

$S = \{HH, HT, TH, TT\}$

$f(z,w)$

	0	1	2
0	0.36	0.24	0
1	0	0.24	0.16

b) $P(\text{at least one head})$

$$= P(1,2) + P(1,1) + P(0,1) \\ = 0.16 + 0.24 + 0.24 \\ = 0.64$$

a) Marginal Distribution of W and of Z

w	0	1	2
$g(w)$	0.36	0.48	0.16

z	0	1
$h(z)$	0.60	0.40

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★ Joint Probability Distribution (for continuous random variable) When X and Y are CRVs, The $f(x, y)$ is a joint density function of X and Y if

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

★ Properties

34 Example: Verify that $f(x, y) = \begin{cases} \frac{2}{5}(2x+3y); & 0 \leq x \leq 1 \\ & 0 \leq y \leq 1 \\ 0 & \text{; elsewhere} \end{cases}$ 1) $f(x, y) \geq 0$

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= \int_0^1 \int_0^1 \frac{2}{5}(2x+3y) dx dy = \int_0^1 \frac{2}{5} [x^2 + 3xy]_0^1 dy \\ &= \int_0^1 \frac{2}{5} (1+3y) dy = \frac{2}{5} \left[y + \frac{3}{2} y^2 \right]_0^1 = \frac{2}{5} \left[1 + \frac{3}{2} \right] = \frac{2}{5} + \frac{3}{5} = 1 \end{aligned}$$

Also calculate $P((X, Y) \in A) = P(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2})$

$$\begin{aligned} \int_{\frac{1}{4}}^{\frac{1}{2}} \int_0^{\frac{1}{2}} \frac{2}{5}(2x+3y) dx dy &= \int_{\frac{1}{4}}^{\frac{1}{2}} \frac{2}{5} [x^2 + 3xy]_0^{\frac{1}{2}} dy = \int_{\frac{1}{4}}^{\frac{1}{2}} \frac{2}{5} \left[\frac{1}{4} + \frac{3}{2} y \right] dy \\ &= \int_{\frac{1}{4}}^{\frac{1}{2}} \left(\frac{2}{20} + \frac{3}{5} y \right) dy = \left[\frac{y}{10} + \frac{3y^2}{10} \right]_{\frac{1}{4}}^{\frac{1}{2}} = \frac{1}{10} \left[\left(\frac{1}{2} + \frac{3}{4} \right) - \left(\frac{1}{4} + \frac{3}{16} \right) \right] \\ &= \frac{13}{160} \end{aligned}$$

35 Example: If $f(x, y) = \begin{cases} k(x^2 + y^2); & 30 \leq x \leq 50, \\ & 30 \leq y \leq 50 \\ 0 & \text{; elsewhere} \end{cases}$ is joint probability density function,

a) Find k

$$\begin{aligned} \int_{30}^{50} \int_{30}^{50} k(x^2 + y^2) dx dy &= 1; \quad k \int_{30}^{50} \left[\frac{x^3}{3} + xy^2 \right]_{30}^{50} dy \\ &= k \int_{30}^{50} \left[\left(\frac{50^3}{3} + 50y^2 \right) - \left(\frac{30^3}{3} + 30y^2 \right) \right] dy = k \int_{30}^{50} \left(20x^2 + \frac{98000}{3} \right) dy \\ &= k \left[\frac{20x^3}{3} + \frac{98000}{3} x \right]_{30}^{50} = k \left[\left(\frac{20}{3}(50^3) + \frac{98000}{3} \cdot 50 \right) - \left(\frac{20}{3}(30^3) + \frac{98000}{3} \cdot 30 \right) \right] \\ &= k \left(\frac{20(98000)}{3} + \frac{98000}{3} \cdot 20 \right) = k \cdot \frac{3920000}{3} = 1 \Rightarrow \boxed{k = \frac{3}{3,920,000}} \end{aligned}$$

b) $P(30 \leq X \leq 40, 40 \leq Y < 50)$

$$\begin{aligned} P &= \int_{30}^{40} \int_{40}^{50} \frac{3}{3920000} (x^2 + y^2) dy dx = \int_{30}^{40} \frac{3}{3,920,000} \left[\frac{x^2 y}{1} + \frac{y^3}{3} \right]_{40}^{50} dx \\ &= \frac{3}{3920000} \left[\frac{10x^3}{3} + \frac{61000}{3} x \right]_{30}^{40} = \frac{3}{3920000} \left[\frac{10}{3}(37000) + \frac{61000}{3} \right] = \frac{3 \cdot 980000}{3920000 \cdot 3} = \frac{1}{4} \end{aligned}$$

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c) Probability both tires are underfilled — means pressure < 40 psi

$$P(X < 40, Y < 40) = \int_{30}^{40} \int_{30}^{40} \frac{3}{3920,000} (x^2 + y^2) dy dx = \frac{37}{196} \approx 0.188$$

36 Example: $f(x, y) = \begin{cases} \frac{1}{2}y & ; 0 < x < y < 1 \\ 0 & ; \text{elsewhere} \end{cases}$ (Find $P(X+Y > \frac{1}{2})$)

Condition $\rightarrow x + y > \frac{1}{2}$ — (1)
 (2)

from 2 $\Rightarrow x > \frac{1}{2} - y$

from 1 $\Rightarrow x < y$

Sol: I don't know 🤔

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- Prob -

★ **Conditional Probability Distribution** - Let X and Y be two random variables, Discrete or Continuous. The conditional distribution of

Random variable Y given that $X = x$ is $f(y/x) = \frac{f(x,y)}{g(x)}$; $g(x) > 0$.

Similarly for X given that $Y = y$ is $f(x/y) = \frac{f(x,y)}{h(y)}$; $h(y) > 0$.

Probability for DRV, lies between a and b when $Y = y$

$$P(a < X < b | Y = y) = \sum_{a < x < b} f(x/y)$$

When X & Y are Continuous and summation extends over all values of X between a & b

$$P(a < X < b | Y = y) = \int_a^b f(x/y) dx$$

37 **Example:** Referring to Example 23. Find conditional distribution of X given that $Y = 1$ and use it to determine $P(X=0 | Y=1)$

$$f(x/1) = \frac{f(x,1)}{h(1)} = \frac{7}{3} f(x,1), \quad x = 0, 1, 2$$

$$h(1) = \sum_{x=0}^2 f(x,1) = f(0,1) + f(1,1) + f(2,1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}$$

$$f(0/1) = \frac{7}{3} \cdot \frac{3}{14} = \frac{1}{2}$$

$$f(1/1) = \frac{7}{3} \cdot \frac{3}{14} = \frac{1}{2}$$

$$f(2/1) = \frac{7}{3} \cdot 0 = 0$$

x	0	1	2
$f(x/1)$	$\frac{1}{2}$	$\frac{1}{2}$	0

$$P(X=0 | Y=1)$$

$$= f(0/1) = \frac{1}{2}$$

Conditional Distribution of X ,
given that $Y = 1$

38 **Example:**

$$f(x,y) = \begin{cases} 10xy^2 & ; 0 < x < y < 1 \\ 0 & ; \text{elsewhere} \end{cases}$$

a) Find marginal $g(x)$, $h(y)$, and conditional density.

$$g(x) = \int_x^1 10xy^2 dy = \frac{10x}{3} y^3 \Big|_x^1 = \frac{10}{3} x(1-x^3)$$

$$h(y) = \int_0^y 10xy^2 dx = \frac{10y^2}{2} x^2 \Big|_0^y = 5y^4$$

$$f(y/x) = \frac{f(x,y)}{g(x)} = \frac{10xy^2}{\frac{10}{3} x(1-x^3)} = \frac{3y^2}{1-x^3}$$

b) $P(Y > \frac{1}{2} | X = \frac{1}{4})$

$$= \int_{1/2}^1 f(y/x = \frac{1}{4}) dy$$

$$= \int_{1/2}^1 \frac{3y^2}{1-(0.25)^3} dy$$

$$= 8/9$$

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(👉👉) NOTE

Please review the examples provided for each topic, for clear understanding of the concepts.

However, the topic of cumulative frequency can be a little confusing. Therefore, I have included the following reference videos:

For Cumulative Frequency for Continuous Random Variables :
https://youtu.be/M5yYo3pJv8M?si=mLmX_VmLOc_rrKb9

For Cumulative Frequency for Discrete Random Variables :
https://youtu.be/IUSDu_RnDno?si=oQOFs58OCOKJTU7c